



Safety Verification of Chiron IR using Constrained Horn Clauses

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Introduction

Program verification provides mathematical guarantees that software behaves correctly, unlike testing, which can find bugs but never prove their absence. While the Chiron framework offers support for dynamic analysis techniques such as fuzzing, symbolic execution, and spectrum-based fault localization, it currently lacks the capability to *prove* that a program satisfies a given specification for *all* possible inputs.

This project bridges the gap between testing and verification in Chiron by implementing a Property Directed Reachability (PDR)-based safety verification engine, that given a program and a set of properties on the program state, either guarantees that they hold or provides a counterexample for the same. We target properties natural to turtle graphics programs, such as spatial bounds on the turtle's position and invariants over program variables.

Base Flow & Tools Used

Overview of the pipeline

The pipeline takes a source turtle program and a set of safety properties as input. Using Chiron's API, the program is translated to Chiron-IR and its relevant user and state variables are extracted. The program semantics are then encoded as Constrained Horn Clauses (CHCs), and combined with the properties expressed in CHC form. This CHC system is given to Z3's **SPACER** engine, which checks whether an **unsafe** state is reachable. A **sat** result yields a concrete counterexample, while **unsat** proves that the property holds.

Features

Category	Option	Description
Mode	universal	Checks the property over all possible values of the user input variables, with the turtle initially at the origin and pendown=True .
	default	Checks the property with all user variables initialized to 0, with the turtle initially at the origin and pendown=False .
	specific	Similar to default , except that the user may explicitly assign values to some or all user input variables.
Property Scope	all	Interprets the safety guarantee as required to hold at every point during program execution.
	terminating	Interprets the safety guarantee as required to hold only at program termination.

Table 1. Verification modes and property scopes supported by the verifier.

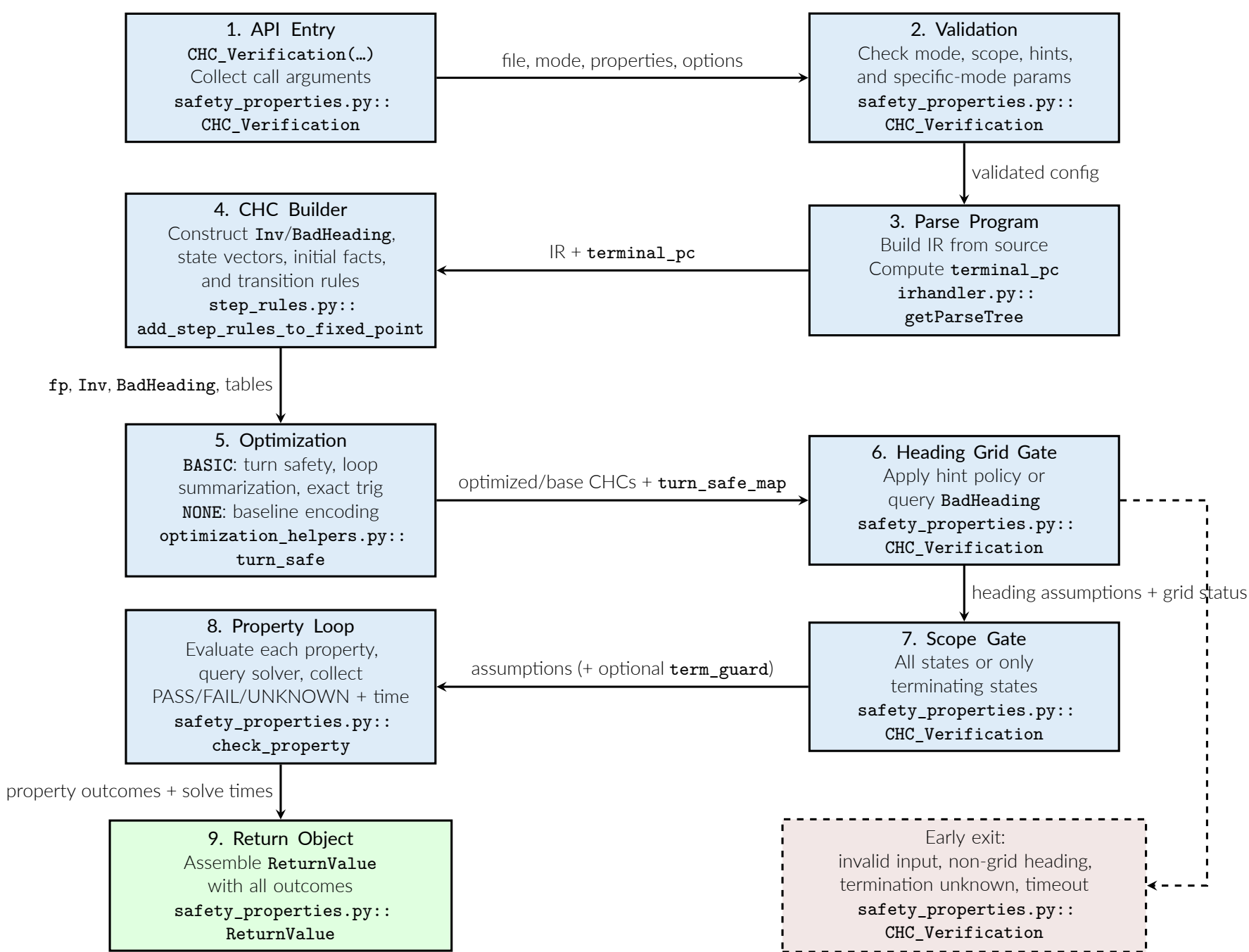


Figure 1. Main verification pipeline showing stage responsibilities and data artifacts flowing through parsing, CHC construction, heading/scope controls, and final result aggregation.

Design Choices & Optimizations

Design Choices

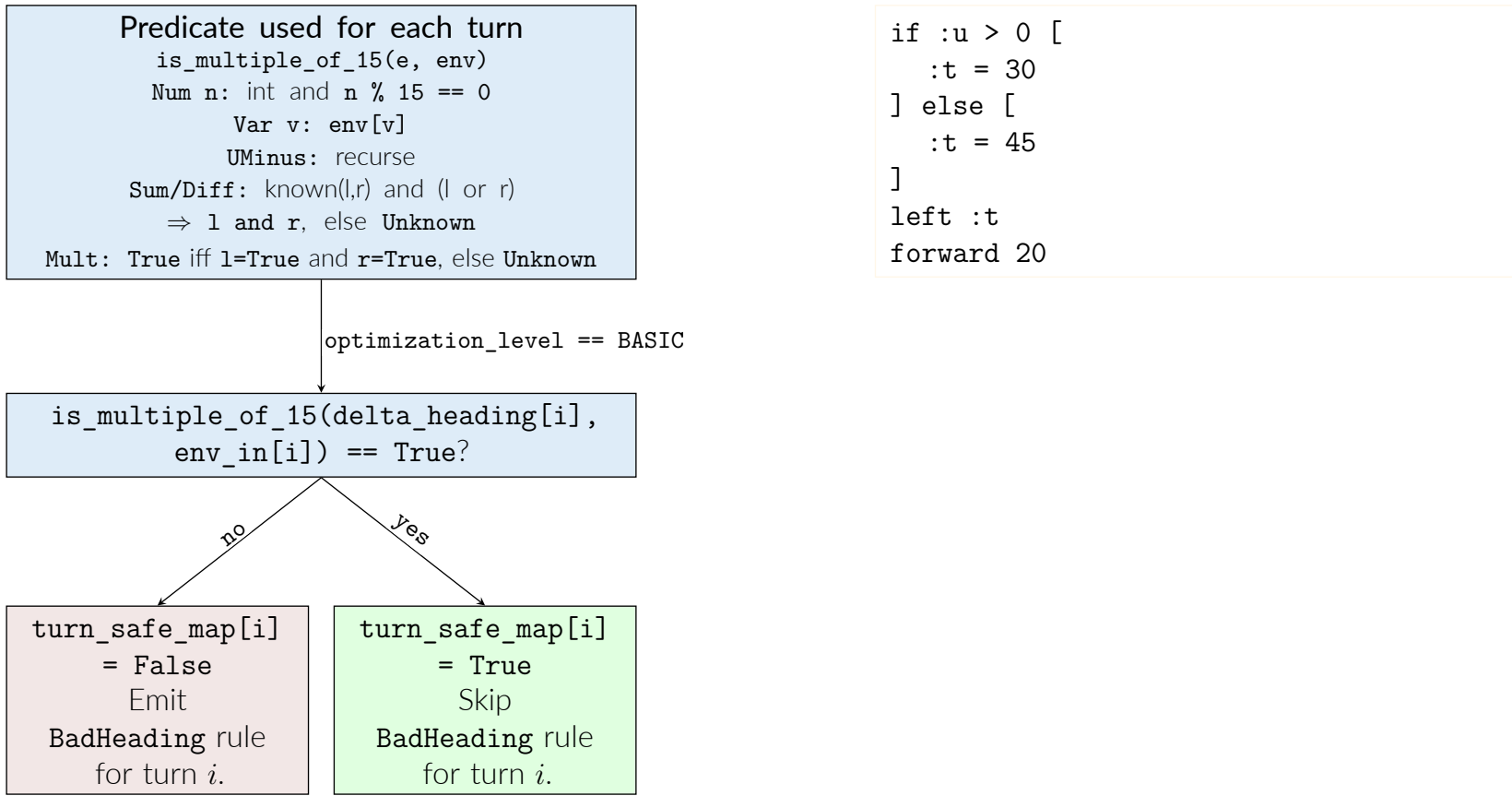
This section summarizes the principal modeling and solver choices adopted in the verifier.

- **CHC semantics with Z3 Fixedpoint/SPACER:** Program behavior is encoded as CHCs and solved using Z3 **Fixedpoint**, with **engine='spacer'** (an IC3/PDR-style Horn-clause engine), with safety posed as **Exists(s, Inv(s) & Not(P(s))**.
- **Single global invariant schema:** A unified relation **Inv(pc, xcor, ycor, heading, pendown, user_vars, loop_counters)** is used across initialization, transition rules, and property queries.
- **Heading-lattice geometry model:** Heading is normalized to **[0,360)** and constrained to **{0,15,...,345}**, with violations represented explicitly via **BadHeading**.

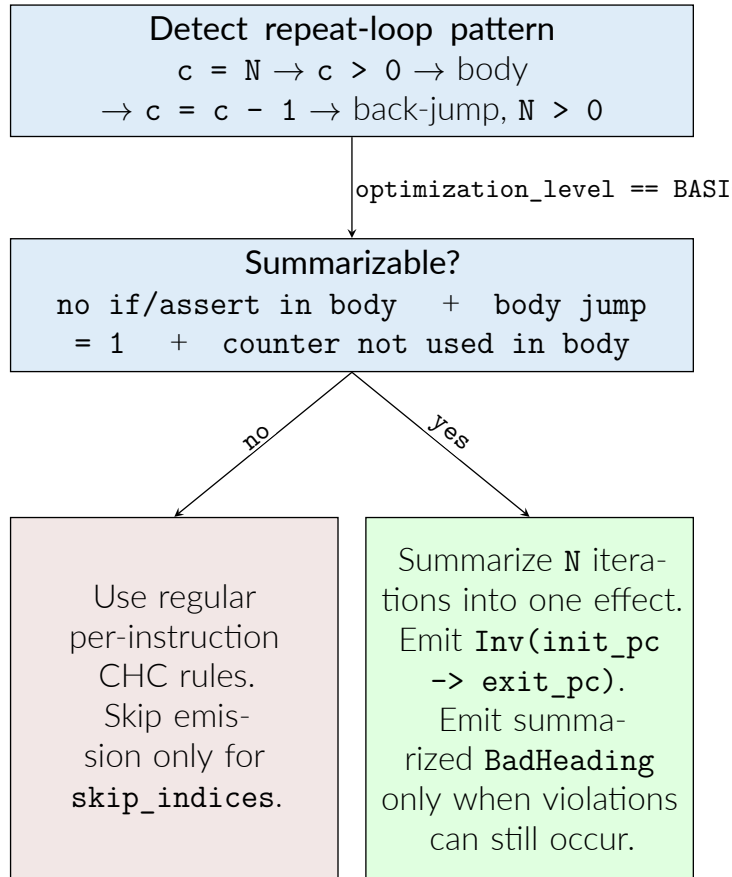
Design Choice	Benefit	Tradeoff
CHCs solved by Z3 SPACER	Supports unbounded safety checking through inductive invariant synthesis and arithmetic obligations; solver outcomes may return counterexample witnesses on violation.	Performance remains sensitive to nonlinear full-arity reasoning, increasing solver cost.
Full state (pc+counters+vars)	Encodes control-state and data-state dependencies in one relation; pc for control/termination, xcor/ycor/heading for geometric predicates, and loop counters for repeat constructs.	Still incur full-arity reasoning, increasing solver cost.
Strict heading lattice + BadHeading	Avoids direct transcendental sin/cos computations; Lookup expansion must remain bounded straints in SPACER by using a finite degree trigonometric lookup table (precomputed in Python) with explicit BadHeading obligations.	Lookup expansion must remain bounded to control CHC size; failure to prove grid preservation yields conservative UNKNOWN outcomes.

Optimizations

Turn-Safe Heading Inference



Repeat-Loop Summarization



Outputs & Perf Analysis

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Bar	2.17	1,392	β
Baz	3.14	83,742	δ
Qux	7.59	974	γ

Table 2. A table caption.

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Future Work

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References

[1] Claude E. Shannon. A mathematical theory of communication. *The Bell System Technical Journal*, 27(3):379–423, 1948. doi:10.1002/j.1538-7305.1948.tb01338.x.